



**A Level • Cambridge (CIE) • Chemistry**

26 mins

14 questions

Multiple Choice Practice

# Topical Practice MCQ

## 17. Carbonyl compounds

Carefully selected past paper questions by topic.

- 33** In which pair will each compound give a different visible result with alkaline  $I_2(aq)$ ?
- A**  $CH_3CH_2OH$  and  $CH_3CHO$
  - B**  $CH_3CHO$  and  $CH_3COCH_3$
  - C**  $CH_3COOH$  and  $C_2H_5COC_2H_5$
  - D**  $CH_3CH_2OH$  and  $C_2H_5CHO$

**34** Structural and stereoisomerism should be considered when answering this question.

P has molecular formula  $C_5H_{10}O$ .

P produces an orange precipitate with 2,4-dinitrophenylhydrazine (2,4-DNPH reagent).

How many isomeric structures does P have?

**A** 5

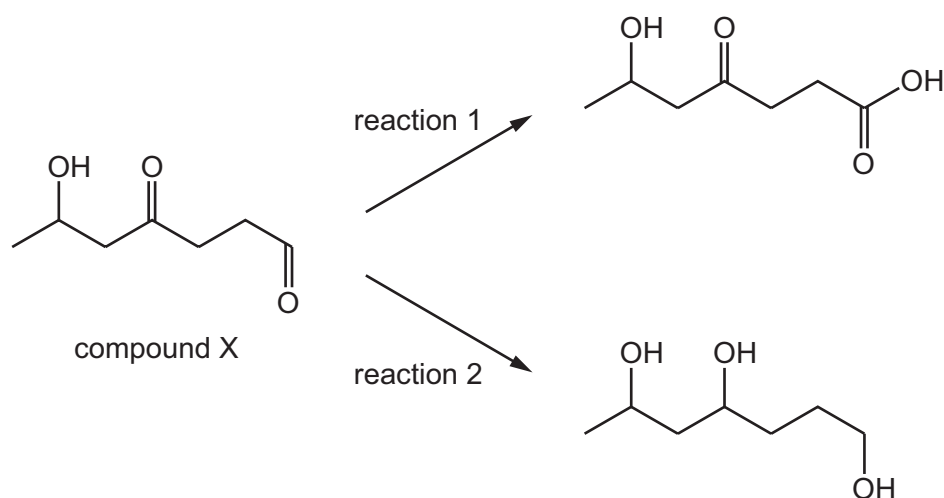
**B** 6

**C** 7

**D** 8

- 35** Two samples of compound X were treated separately with different reagents which were added in excess.

The products of these two reactions are shown.



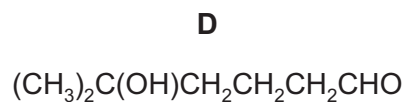
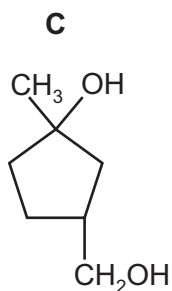
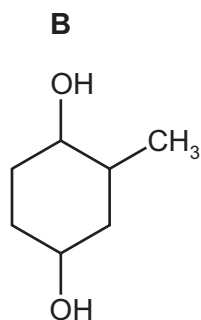
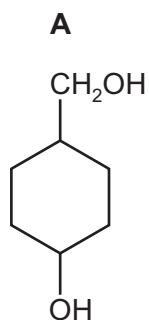
Which reagents could be used for reaction 1 and reaction 2?

|          | reaction 1                           | reaction 2        |
|----------|--------------------------------------|-------------------|
| <b>A</b> | hot acidified sodium dichromate(VI)  | Na                |
| <b>B</b> | hot acidified sodium dichromate(VI)  | NaBH <sub>4</sub> |
| <b>C</b> | Tollens' reagent followed by HCl(aq) | Na                |
| <b>D</b> | Tollens' reagent followed by HCl(aq) | NaBH <sub>4</sub> |

- 35** Heating compound X,  $C_7H_{14}O_2$ , under reflux with an excess of acidified potassium dichromate(VI) produces compound Y.

Compound Y produces hydrogen gas with sodium metal and forms orange crystals with 2,4-DNPH reagent.

What could X be?

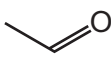
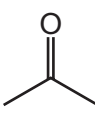
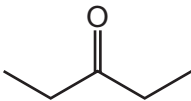


- 38** Compound Y is heated with a mild oxidising agent. One of the products of the reaction reacts with hydrogen cyanide forming 2-hydroxybutanenitrile.

What is compound Y?

- A** butan-1-ol
- B** butan-2-ol
- C** propan-1-ol
- D** propan-2-ol

35 The table shows a student's predictions for the reactions of three compounds.

|   | compound                                                                          | alkaline<br>$I_2(aq)$ | Fehling's<br>reagent | Tollens'<br>reagent |
|---|-----------------------------------------------------------------------------------|-----------------------|----------------------|---------------------|
| 1 |  | ✓                     | ✓                    | ✓                   |
| 2 |  | ✓                     | x                    | x                   |
| 3 |  | x                     | x                    | x                   |

key

✓ = reaction occurs

x = no reaction

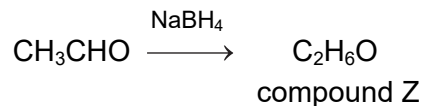
Which rows show the correct predictions?

- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

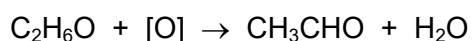
**27** In this question you can assume that  $^1\text{H}$  and  $^3\text{H}$  have the same chemical properties.

A sample of ethanal contains only one isotope of hydrogen,  $^1\text{H}$ .

It is reduced to compound Z,  $\text{C}_2\text{H}_6\text{O}$ , in a nucleophilic addition reaction using  $\text{NaBH}_4$ . All the hydrogen atoms in the  $\text{NaBH}_4$  are the  $^3\text{H}$  isotope.



Compound Z is then oxidised back to ethanal and water.



Which statement about the final mixture of products is correct?

- A** Both ethanal and water contain  $^3\text{H}$  atoms.
- B** Ethanal is the only product containing  $^3\text{H}$  atoms.
- C** Neither ethanal nor water contain  $^3\text{H}$  atoms.
- D** Water is the only product containing  $^3\text{H}$  atoms.



**27** How many structural isomers are there of molecular formula  $\text{C}_5\text{H}_{10}\text{O}$  that give a red precipitate with Fehling's solution?

**A** 1

**B** 2

**C** 3

**D** 4

**28** Structural isomerism **only** should be considered when answering this question.

A set of isomeric compounds, with molecular formula  $C_5H_{10}O$ , all react in a 1:1 ratio with an excess of HCN by nucleophilic addition.

How many isomeric compounds are in the set?

**A** 5

**B** 6

**C** 7

**D** 8

- 27** Compound **X** is heated with a mild oxidising agent. One of the products of the reaction will react with hydrogen cyanide, forming 2-hydroxybutanenitrile.

What is compound **X**?

- A** butan-1-ol
- B** butan-2-ol
- C** propan-1-ol
- D** propan-2-ol

26 Hept-4-enal is present in cow's milk.

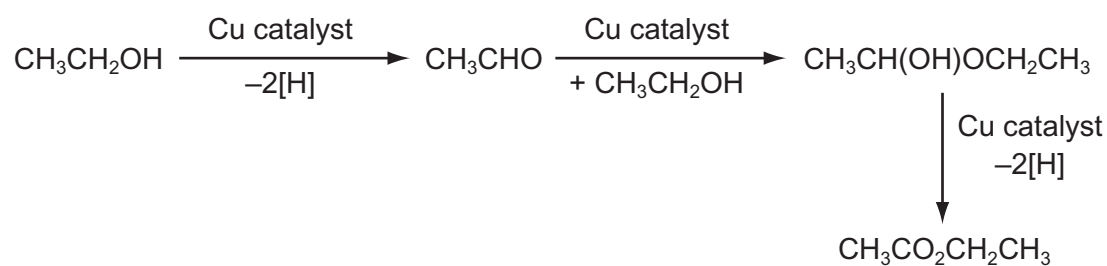


hept-4-enal

Which row correctly shows the product formed when hept-4-enal is treated with the given reducing agent?

|          | reducing agent           | product                                          |
|----------|--------------------------|--------------------------------------------------|
| <b>A</b> | $\text{H}_2 + \text{Ni}$ | $\text{CH}_3(\text{CH}_2)_5\text{CH}_2\text{OH}$ |
| <b>B</b> | $\text{H}_2 + \text{Ni}$ | $\text{CH}_3(\text{CH}_2)_5\text{CH}_3$          |
| <b>C</b> | $\text{NaBH}_4$          | $\text{CH}_3(\text{CH}_2)_5\text{CH}_2\text{OH}$ |
| <b>D</b> | $\text{NaBH}_4$          | $\text{CH}_3(\text{CH}_2)_5\text{CHO}$           |

- 20 A new industrial preparation of ethyl ethanoate has been developed using cheap sources of ethanol.

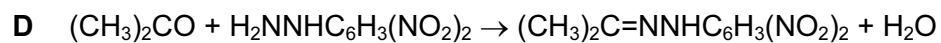
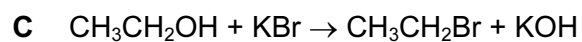
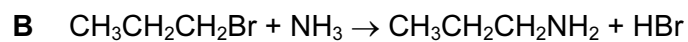
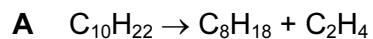


Which process is involved at some stage in this reaction sequence?

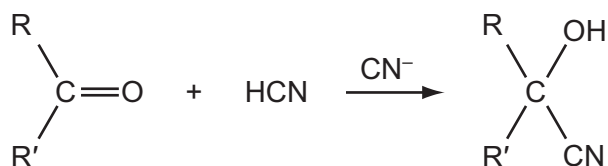
- A electrophilic addition
- B nucleophilic addition
- C nucleophilic substitution
- D reduction

25 Many, but not all, organic reactions need to be heated before reaction occurs.

Which reaction occurs at a good rate at room temperature (20 °C)?



- 27 Cyanohydrins can be made from carbonyl compounds by generating  $\text{CN}^-$  ions from  $\text{HCN}$  in the presence of a weak base.



In a similar reaction,  $^-\text{CH}_2\text{CO}_2\text{CH}_3$  ions are generated from  $\text{CH}_3\text{CO}_2\text{CH}_3$  by strong bases.

Which compound can be made from an aldehyde and  $\text{CH}_3\text{CO}_2\text{CH}_3$  in the presence of a strong base?

- A  $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{CH}_3$
- B  $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- C  $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CO}_2\text{CH}_3$
- D  $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CO}_2\text{CH}_3$

## Answer Key

| No. | Paper          | Topic                  | Answer |
|-----|----------------|------------------------|--------|
| 1   | 9701_m24_qp_12 | 17. Carbonyl compounds | D      |
| 2   | 9701_m23_qp_12 | 17. Carbonyl compounds | D      |
| 3   | 9701_s23_qp_11 | 17. Carbonyl compounds | D      |
| 4   | 9701_w23_qp_12 | 17. Carbonyl compounds | A      |
| 5   | 9701_m22_qp_12 | 17. Carbonyl compounds | C      |
| 6   | 9701_w22_qp_12 | 17. Carbonyl compounds | A      |
| 7   | 9701_m21_qp_12 | 17. Carbonyl compounds | A      |
| 8   | 9701_w20_qp_12 | 17. Carbonyl compounds | D      |
| 9   | 9701_s19_qp_13 | 17. Carbonyl compounds | C      |
| 10  | 9701_w16_qp_12 | 17. Carbonyl compounds | C      |
| 11  | 9701_s14_qp_13 | 17. Carbonyl compounds | A      |
| 12  | 9701_w13_qp_12 | 17. Carbonyl compounds | B      |
| 13  | 9701_w10_qp_12 | 17. Carbonyl compounds | D      |
| 14  | 9701_w09_qp_12 | 17. Carbonyl compounds | C      |